Clean Pro Model Response

Architecture Design for WiFi CSI Gait Recognition

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I'll treat this as a reviewer-facing architecture design task: verify prior CSI/WiFi sensing models, reject superficial model-swapping, and focus on defensible Q1-level blueprints.

Recent literature already covers many obvious baselines, including direction handling, residual-CNN Transformers, cross-domain gait adaptation, and multi-view CSI. I'll therefore prioritize CSI-specific factorization, condition/domain modeling, hard protocols, and self-supervised pretraining.

Executive verdict

Your dataset can support architecture-level Q1 papers, but only if you avoid the easy trap: do not publish "CNN/LSTM/Transformer reaches high accuracy on 25 people." That is now weak. Recent CSI sensing literature already contains CNNs, CNN-LSTMs, CNN-BiGRUs, ViTs, generic Transformers, self-supervised learning, few-shot open-set recognition, and domain-adaptation methods. Your defensible novelty should come from architectures that explicitly exploit your dataset's structure:

identity × direction × bag condition × regular/modified router setup × paired same-subject cross-setup samples.

Your dataset has: 25 regular-router participants with 100 samples each; 5 modified-router participants with 100 samples each; the 5 modified-router participants also exist in the regular-router setup; odd/even sample index gives walking direction; samples 1–80 are normal walking; samples 81–100 are with-bag walking.

The best Q1 architecture direction is:

Paper 1: condition-disentangled, direction-equivariant, identity-preserving CSI representation learning on the 25-person regular-router dataset.

Paper 2: paired regular-to-modified antenna-setup domain adaptation using the 5 overlapping subjects, framed conservatively as a cross-configuration pilot unless you collect more modified-router subjects.

PART 1 — Current literature verification

1.1 What has already been used

The field has moved beyond simple classifiers. Early works used handcrafted gait features and SVM-like classifiers; later works used deep CNNs, CNN-LSTM, CNN-BiGRU, Transformers, ViT-style CSI images, few-shot learning, GAN/domain adaptation, and self-supervised learning.

Table 1a — columns: Dataset, Task, Model used, Input representation, Key contribution

Paper / year	Dataset	Task	Model used	Input representation	Key contribution
WiWho, 2016	20 volunteers, multiple locations	WiFi gait-based person ID	Handcrafted gait/CSI features	CSI perturbation during walking	Early proof that WiFi CSI gait can identify people; reported accuracy decreased from about 92% to 80% for group sizes 2–6
WiFi-ID, 2016	COTS WiFi setup	Human identification	Fourier/feature selection + SVM-style ML	CSI features	Showed WiFi CSI can capture gait-induced identity signatures
NeuralWave, 2018	Private dataset, 24 people	Gait-based user ID	23-layer CNN / RadioNet	Sanitized CSI / spectrogram-like representation	Early deep learning for CSI gait biometrics; reported 87.76 ± 2.14% for 24 people
CSIID, 2019	COTS WiFi	Human identification	CNN + LSTM	CSI time-series	Combined convolutional and recurrent layers for gait feature extraction
Gate-ID, 2020	20 subjects crossing back/forth	Direction-invariant human ID	Direction estimation + augmentation/attention-style model	CSI spatial-temporal features	Explicitly studied mirror-like CSI patterns caused by walking direction and antenna orientation
CAUTION, 2022	WiFi gait authentication	Few-shot open-set authentication	Few-shot user profiles + intruder threshold	CSI gait features	Handles limited enrollment and unknown intruders
AutoFi, 2022	Atheros WiFi + public datasets	Automatic WiFi sensing; includes gait transfer	Geometric self-supervised learning	Unlabeled CSI samples	Annotation-efficient, cross-task self-supervised WiFi sensing
SenseFi, 2023	Multiple public CSI datasets	WiFi sensing benchmark	MLP, CNN, RNN, Transformers	CSI tensors	Benchmark/library for deep WiFi sensing
Wi-Gait, 2023	CSI gait/person ID	Robust personnel identification	Multiscale residual convolution-enhanced Transformer	WiFi CSI gait features	Transformer + multiscale residual convolution for robust gait ID
DCS-Gait, 2024	34 subjects, 2 scenes, 3 states	Cross-scene/cross-state gait recognition	Class-level domain adaptation	WiFi CSI	Strong cross-domain WiFi gait adaptation
MaP-SGAN, 2024	Cross-domain CSI gait	Cross-domain gait recognition	Multi-anchor point Siamese GAN	WiFi CSI	Addresses asymmetric cross-domain gait recognition
NeuralWiGait, 2024/2025	Widar 3.0 and NTU-Fi-HumanID	WiFi gait recognition	CNN + BiGRU	CSI samples	Hybrid CNN-BiGRU with high reported accuracy
ViT for WiFi CSI HAR, 2024	UT-HAR, NTU-Fi HAR	Activity recognition	ViT, SimpleViT, DeepViT, Swin, CaiT	CSI spectra / image-like forms	Demonstrates ViT family on CSI HAR
WiSigPro, 2024	CSI HAR	Activity recognition	Transformer with attention and signal processing	CSI spatiotemporal data	Transformer for passive WiFi sensing
Multi-view CSI gait fusion, 2025/2026	CSI gait recognition	Gait recognition with few-sample adaptability	GADF, subcarrier diversity heatmaps, ViT, CNN+Transformer fusion	Raw CSI + image-like views	Multi-view feature extraction and fusion for CSI gait
Transformer-based CSI amplitude/phase person ID, 2025	ESP32 dataset, 6 participants	Stationary person ID	Dual-branch Transformer	Amplitude + phase	Separate amplitude/phase Transformer branches

Paper / year	Dataset	Task	Model used	Input representation	Key contribution
CSI-Bench, 2025	35 users, 26 environments, 16 device types, 460+ hours	Multitask WiFi sensing benchmark	Standardized supervised, multitask, few-shot baselines	Commercial WiFi CSI	Raises the bar for generalization and reproducibility

Table 1b — columns: Limitations, Whether your proposed direction would be novel

Paper / year	Limitations	Whether your proposed direction would be novel
WiWho, 2016	Small closed-set groups; no modern deep robustness; limited covariate modeling	Your direction is novel if you focus on direction/bag/domain robustness, not just ID accuracy.
WiFi-ID, 2016	Classical ML; limited scalability; weak robustness protocols	Your deep factorized architecture is novel relative to this, but SVM baselines are mandatory.
NeuralWave, 2018	CNN-only; little nuisance disentanglement; no bag/setup adaptation	Plain CNN is not novel; physically/condition-aware CNN variants could be.
CSIID, 2019	CNN-LSTM is now overused	Do not claim CNN-LSTM novelty; use only as baseline.
Gate-ID, 2020	Direction addressed, but not bag/no-bag, antenna-spread setup, paired cross-setup transfer	Your direction-aware model must beat/extend this idea: direction equivariance + bag robustness + setup transfer.
CAUTION, 2022	Not focused on physical antenna setup, bag condition, or direction-factor disentanglement	Few-shot/open-set extensions are Q1-relevant but must be redesigned for your covariates.
AutoFi, 2022	Not designed around identity/direction/bag/setup factorization	Masked/self-supervised CSI pretraining is publishable if linked to your hard protocols.
SenseFi, 2023	Shows generic deep models are baseline-level now	You need architecture novelty plus strong splits, not just model zoo results.
Wi-Gait, 2023	Generic Transformer novelty is reduced after this	Any Transformer proposal must add factorization, physics, domain, or self-supervision.
DCS-Gait, 2024	Already covers domain adaptation broadly	Your novelty must be paired router-configuration / antenna-spread adaptation, not generic DA.
MaP-SGAN, 2024	GAN/domain adaptation already used	Avoid “GAN for CSI gait” unless tied to your paired regular-modified setup and strong baselines.
NeuralWiGait, 2024/2025	CNN-BiGRU is now clearly baseline-level	Use as baseline, not contribution.
ViT for WiFi CSI HAR, 2024	HAR, not gait ID; still makes “ViT on CSI image” non-novel	Your ViT must be CSI-specific: direction, bag, setup, amplitude-phase, or physics-aware.
WiSigPro, 2024	Generic attention/Transformer is saturated	Avoid claiming novelty from positional encoding or MHA alone.
Multi-view CSI gait fusion, 2025/2026	Reduces novelty of simple amplitude/spectrogram multi-view fusion	Your multi-view idea must add factorized covariate robustness or domain transfer.
Transformer-based CSI amplitude/phase person ID, 2025	Stationary, small participant count, not gait-focused	Amplitude+phase dual branch alone is not enough; use phase carefully with gait-specific losses.
CSI-Bench, 2025	Much larger than your dataset; not specifically your antenna setup	Your paper must be honest about single-environment limits and emphasize hard internal protocols.

1.2 What is overused

These are baseline-level, not Q1 contributions by themselves:

Overused direction	Reviewer reaction
CNN on CSI images	Too common.
CNN-LSTM / CNN-GRU / CNN-BiGRU	Already used in CSIID and NeuralWiGait-like systems.
Vanilla Transformer encoder	Already used in Wi-Gait, WiSigPro, ViT-HAR, stationary CSI ID.
ViT on CSI spectrogram	Already explored for CSI HAR and now multi-view CSI gait.
Softmax-only person classification	Too shallow for biometric/gait papers.
Random train/test split only	Serious leakage concern.
“Amplitude + phase fusion” without phase sanitization	Weak and risky.
GAN augmentation without hard cross-domain protocol	Superficial.
LLM over CSI converted to text	Gimmicky and scientifically weak.

Data leakage is now a serious reviewer concern in WiFi CSI sensing; recent papers explicitly show inflated CSI-HAR performance caused by improper partitioning. Your architecture must therefore be paired with leakage-safe protocols.

1.3 What is still underexplored for CSI gait recognition

I would not claim these are “never implemented anywhere,” but they appear likely underexplored specifically for CSI-based gait/person recognition:

1. Latent factorization of identity, direction, bag condition, and router setup in one architecture.
2. **Direction equivariance:** modeling left-to-right and right-to-left walking as a learnable latent transformation, not merely as data augmentation.
3. Paired regular-modified router adaptation using the same 5 subjects as a cross-setup bridge.
4. **Antenna-spread physical priors:** treating regular vs 4 m antenna-spread setup as a spatial-aperture shift.
5. Hypergraph modeling of time × subcarrier × antenna-link CSI structure.
6. Masked amplitude-phase CSI pretraining with identity-preserving fine-tuning.
7. **Counterfactual condition invariance:** “same identity under normal/bag and opposite direction.”
8. Open-set biometric verification under direction/bag/setup shift, not just closed-set classification.
9. Router-configuration-conditioned mixture-of-experts.
10. Causal/nuisance deconfounding for CSI biometrics.

PART 2 — Architecture design requirements for your dataset

Your architecture should explicitly exploit these labels:

```
person_id     = folder/person
direction     = odd sample -> L2R, even sample -> R2L
condition     = samples 1–80 normal, 81–100 with bag
setup         = regular or modified
cross_setup   = available for same 5 subjects
```

The architecture should satisfy these reviewer-facing requirements:

Requirement	Why it matters
Identity embedding must be robust to direction	Gate-ID already showed direction changes can produce mirror-like CSI perturbations.
Identity embedding must be robust to bag condition	Your samples 81–100 are valuable because bag carrying is a real covariate.
Direction and bag labels should not merely be ignored	Use them as auxiliary supervision, adversarial nuisance labels, or contrastive pairing.
Regular vs modified setup must be treated as domain shift	The 4 m antenna-spread setup changes spatial sampling and multipath response.
Same-subject cross-setup samples should be used carefully	They enable paired alignment, but only for 5 subjects.
Amplitude and phase should be handled separately first	CSI phase is useful but sensitive to synchronization errors; phase sanitization is mandatory.
Evaluation must include hard splits	Random split alone will look weak or leaky.
Architecture must work with small data	2,500 regular samples and 500 modified samples are not enough for huge foundation models.
The model must include biometric metrics	Accuracy alone is insufficient; include EER, AUROC, TAR@FAR for verification.
Complexity must be reported	Q1 reviewers expect parameters, FLOPs, memory, and inference time.

PART 3 — Type A: completely new architecture ideas for CSI gait recognition

Strict wording: these are apparently not reported as complete CSI-gait architectures in the literature I found. In a manuscript, write “to the best of our knowledge” and include a careful related-work comparison.

A1. CID-GaitNet: Condition-Invariant Disentangled CSI Gait Network

Q1 title:
“Condition-Invariant Disentangled Representation Learning for WiFi CSI Gait Biometrics Under Direction and Carrying Variations”

Core idea:
Learn separate latent subspaces for identity, walking direction, and bag condition. The identity subspace is trained to be discriminative for person ID but adversarially uninformative about direction and bag condition.

Why it is novel:
CSI gait works often use auxiliary labels or augmentation, but a full identity-direction-bag disentanglement architecture for CSI gait recognition appears underexplored. This directly exploits your label design.

Inspired by:
Disentangled representation learning, adversarial nuisance removal, ArcFace biometric embeddings, supervised contrastive learning.

Suitable for your dataset:
Best fit for the 25-person regular-router dataset because you have direction and bag labels for every participant.

Input:
Amplitude or amplitude + sanitized phase CSI tensor:

```
X ∈ ℝ^{B×T×S×L×C}
B = batch
T = packets/time steps
S = subcarriers
L = antenna links / spatial streams
C = modalities: amplitude, phase
```

Pipeline:

```
CSI tensor
→ amplitude/phase preprocessing
→ separable CSI stem
→ temporal-subcarrier encoder
→ shared embedding z
→ identity projector z_id
→ direction projector z_dir
→ condition projector z_bag
→ ArcFace identity head
→ direction auxiliary head
→ bag auxiliary head
→ adversarial nuisance heads on z_id
```

Losses:

```
L = L_arcface
+ λ₁ L_supcon_identity
```

```
+ λ2 L_direction_CE
+ λ3 L_bag_CE
+ λ4 L_adversarial_direction_on_identity
+ λ5 L_adversarial_bag_on_identity
+ λ6 L_orthogonality
+ λ7 L_consistency
```

Training strategy:
Start with amplitude-only. Add sanitized phase after baseline stability. Train with balanced mini-batches: same identity, both directions, normal/bag if possible.

Evaluation protocol:
Random split, grouped split, L2R→R2L, R2L→L2R, normal→bag, normal-L2R→bag-R2L, few-shot, open-set.

Required baselines:
SVM, RF/XGBoost, 1D CNN, CNN-GRU, TCN, Transformer, Wi-Gait-style multiscale Transformer, ArcFace-only model.

Ablations:
Remove adversarial heads, remove orthogonality, remove SupCon, remove direction head, remove bag head, amplitude only vs phase only vs fused.

Expected advantages:
Strongest Paper 1 model. Defensible because it uses labels unique to your dataset.

Weaknesses:
Bag samples are only 20 per subject, so bag robustness may have high variance.

Risk of rejection:
Medium if results only improve random split. Low-to-medium if hard-split gains are strong.

Q1 defense:
“Unlike prior CSI gait models that treat direction and carrying state as nuisance noise or simple augmentation, CID-GaitNet explicitly factorizes identity, direction, and carrying condition and validates the learned invariance under leave-condition-out protocols.”

Best paper:
Paper 1.

A2. CycleDIR-CSI: Direction-Equivariant Cycle Transformer for CSI Gait

Q1 title:
“Learning Direction Equivariance in WiFi CSI Gait Recognition via Latent Cycle Constraints”

Core idea:
Model L2R and R2L walking as a learnable latent transformation. Identity should remain invariant; motion-direction components should transform predictably.

Novelty:
Gate-ID studies direction effects, but this architecture treats opposite walking direction as a mathematical equivariance relation:

```
f_id(x_{L2R}) ≈ f_id(x_{R2L})
f_motion(x_{R2L}) ≈ Q f_motion(x_{L2R})
```

```
Q² ≈ I
```

Inspired by:
Equivariant representation learning, cycle consistency, group theory.

Suitable for your dataset:
Odd/even sample structure gives abundant L2R/R2L samples for every participant and both normal/bag conditions.

Input:
Amplitude CSI or amplitude + phase tensor.

Pipeline:

```
CSI input
→ CSI encoder
→ split latent into identity vector and direction-sensitive vector
→ learnable direction operator Q
→ identity classifier
→ direction classifier
→ cycle-equivalence loss
```

Losses:
ArcFace, direction CE, supervised contrastive identity loss, cycle loss, involution loss $\|Q^2 - I\|$, identity consistency across directions.

Training strategy:
Sample pseudo-pairs: same person, same condition, opposite direction.

Evaluation:
Mainly cross-direction and combined cross-direction + bag.

Baselines:
Gate-ID-style direction augmentation, CNN-GRU, TCN, Transformer, CID-GaitNet without cycle operator.

Ablations:
No Q operator, no cycle loss, no pseudo-pair mining, random vs structured pairing.

Expected advantage:
Very strong direction-robust architecture.

Weakness:
Pairs are not exact repeated walks; they are same-subject/same-condition pseudo-pairs.

Risk:
Medium. Needs careful explanation.

Q1 defense:
“Direction is not treated as a class label only; it is encoded as an equivariant latent transformation with explicit cycle constraints.”

Best paper:
Paper 1.

A3. PAIR-BridgeNet: Paired Antenna-Configuration Invariant Router Bridge Network

Q1 title:
“Paired Cross-Configuration Domain Adaptation for WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Setups”

Core idea:
Use the 5 same-subject regular/modified pairs to learn a bridge between compact-router CSI and antenna-spread-router CSI.

Novelty:
Domain adaptation has been used in CSI gait, but your regular-to-modified router configuration is a specific physical setup shift. The paired same-subject subset gives a unique alignment signal.

Inspired by:
DANN, MMD/CORAL, cycle translation, prototype alignment, teacher-student learning.

Suitable for your dataset:
Directly uses the 5 participants who exist in both setup types.

Input:
Regular and modified CSI tensors.

Pipeline:

```
regular CSI branch      modified CSI branch
↓                      ↓
domain-specific stem     domain-specific stem
↓                      ↓
shared CSI encoder
↓
identity embedding
↓
prototype memory
↓
domain adversarial head + pair alignment + identity head
```

Losses:
ArcFace, supervised contrastive, DANN domain loss, MMD/CORAL, pair alignment loss, cycle adapter loss, prototype consistency.

Training:
Pretrain encoder on 25 regular participants. Fine-tune/adapt using 5 overlapping regular/modified subjects.

Evaluation:
R→M, M→R, few-shot M adaptation, no-adaptation vs adaptation.

Baselines:
No adaptation, fine-tuning, CORAL, MMD, DANN, MaP-SGAN-style baseline if feasible, DCS-Gait-inspired class-level DA.

Ablations:
No pair loss, no domain adversarial loss, no prototype memory, no setup-specific stem, no pretraining.

Expected advantage:
Best Paper 2 architecture.

Weakness:
Only 5 modified subjects. Cannot claim broad deployment.

Risk:
High if overclaimed; medium if framed as paired cross-configuration study.

Q1 defense:
“Unlike generic cross-scene DA, this work studies a controlled physical antenna-configuration shift with paired same-subject CSI.”

Best paper:
Paper 2.

A4. Fresnel-HyperGait: Physics-Guided Hypergraph Transformer for CSI Gait

Q1 title:
“A Physics-Guided Hypergraph Transformer for WiFi CSI Gait Recognition With Antenna-Spread Spatial Priors”

Core idea:
Represent CSI as a hypergraph over time patches, subcarrier groups, antenna links, and setup geometry. Hyperedges encode physical relationships: same time, same subcarrier band, same antenna link, and setup-specific spatial aperture.

Novelty:
GNNs and Transformers exist, but a CSI-gait-specific hypergraph encoding of time × subcarrier × antenna-link × router geometry is likely underexplored.

Inspired by:
Hypergraph neural networks, graph Transformers, RF/Fresnel-zone physical priors.

Suitable for your dataset:
Especially useful for comparing regular and antenna-spread setups.

Input:
CSI tensor plus metadata tokens for setup type and antenna-link index.

Pipeline:

```
CSI tensor
→ patch embedding
→ node construction: time-subcarrier-link tokens
→ hyperedge construction
→ hypergraph attention layers
→ temporal pooling
→ identity / direction / bag / setup heads
```

Losses:
ArcFace, SupCon, domain adversarial, graph smoothness, subcarrier-link dropout consistency.

Training:
Pretrain on regular data; adapt with modified data.

Evaluation:
Person ID, setup shift, antenna-link ablation.

Baselines:
Transformer without graph, GCN, GAT, TCN, Wi-Gait-style Transformer.

Ablations:
Remove hyperedges type-by-type; remove geometry encoding; random graph vs physics graph.

Expected advantage:
Strong architectural novelty.

Weakness:
Requires exact antenna-link metadata and stable CSI dimensions.

Risk:
Medium-high. Reviewers may challenge physical assumptions.

Q1 defense:
“CSI is not a generic image; it is a structured RF tensor. The model encodes known RF acquisition axes explicitly.”

Best paper:
Paper 2 or advanced Paper 1 extension.

A5. M2CSI-MAE: Masked Multi-Modal CSI Autoencoding Transformer

Q1 title:
“Masked Multi-Modal CSI Modeling for Few-Shot WiFi Gait-Based Person Recognition”

Core idea:
Pretrain by masking packets, subcarriers, antenna links, amplitude, and phase, then reconstruct missing CSI. Fine-tune for identity using ArcFace/SupCon.

Novelty:
Masked time-series autoencoders exist, and AutoFi uses self-supervision, but a CSI gait-specific amplitude-phase, subcarrier-link, condition-aware MAE is still strong if designed carefully. Masked time-series autoencoders are a serious alternative to calling this an LLM.

Input:
Amplitude + sanitized phase.

Pipeline:

```
CSI tensor
→ random structured mask
→ visible-token encoder
→ lightweight decoder
→ reconstruct amplitude, phase, temporal derivative, Doppler/spectrogram view
→ fine-tune identity head
```

Losses:
Masked reconstruction, phase derivative loss, contrastive identity loss, ArcFace after fine-tuning.

Training:
Use all regular + modified data for self-supervised pretraining; supervised fine-tune on regular 25-person labels.

Evaluation:
5-shot, 10-shot, 20-shot, normal→bag, L2R→R2L.

Baselines:
Supervised Transformer, SimCLR, TS2Vec, AutoFi-inspired SSL, MAE without phase.

Ablations:
Mask ratio, mask type, amplitude-only, phase-only, no reconstruction pretraining, no contrastive fine-tuning.

Expected advantage:
Strong if labeled data is limited.

Weakness:
May not beat supervised training if dataset is clean and not too small.

Risk:
Medium.

Q1 defense:
“Pretraining is not generic; masking is aligned with CSI acquisition axes and downstream biometric invariance.”

Best paper:
Paper 1 or separate SSL paper.

A6. ProtoRouter-Gait: Router-Aware Prototype Memory Network

Q1 title:
“Router-Aware Prototype Memory for Few-Shot and Cross-Setup WiFi CSI Gait Recognition”

Core idea:
Maintain identity prototypes for each person and condition:

```
P_id
P_id,direction
P_id,condition
P_id,setup
```

Then learn adapters that map direction/bag/setup-specific prototypes into a canonical identity prototype.

Novelty:
Few-shot CSI authentication exists, but condition-specific prototype memory for identity × direction × bag × router setup is likely underexplored.

Input:
CSI embedding from CNN/TCN/Transformer encoder.

Pipeline:

```
CSI encoder
```

```
  → embedding
  → prototype memory lookup/update
  → condition adapter
  → metric classifier
  → open-set threshold
```

Losses:
Prototype loss, ArcFace/CosFace, triplet, open-set margin, condition consistency.

Training:
Episodic training: N-way K-shot tasks with support/query splits.

Evaluation:
5-shot, 10-shot, 20-shot; open-set impostor rejection; cross-bag and cross-direction few-shot.

Baselines:
ProtoNet, MatchingNet, RelationNet, CAUTION-style profile threshold, triplet/Siamese.

Ablations:
No condition prototype, no setup prototype, Euclidean vs cosine, fixed vs learnable prototypes.

Expected advantage:
Excellent for biometric framing.

Weakness:
Not as strong for pure closed-set classification.

Risk:
Medium.

Q1 defense:
“Transforms CSI gait recognition from closed-set classification to realistic enrollment-based biometrics.”

Best paper:
Paper 1.

A7. CSI-CausalGait: Causal Nuisance Deconfounding Network

Q1 title:
“Causal Nuisance Deconfounding for Robust WiFi CSI Gait Biometrics”

Core idea:
Treat identity as the target causal factor and direction, bag, and setup as observed nuisance variables. Learn identity representation invariant to interventions on nuisance factors.

Novelty:
Causal invariance is rare in CSI gait recognition.

Input:
Amplitude or amplitude+phase CSI.

Pipeline:

```
CSI encoder
→ latent factors
→ identity predictor
→ nuisance predictors
→ adversarial deconfounding
→ counterfactual consistency module
```

Losses:
ArcFace, invariance risk minimization-like loss, adversarial nuisance loss, counterfactual consistency, SupCon.

Training:
Treat each condition/domain as an environment:

```
E1 = normal L2R
E2 = normal R2L
E3 = bag L2R
E4 = bag R2L
E5 = modified setup
```

Evaluation:
Leave-one-environment-out.

Baselines:
ERM, GroupDRO, IRM-style baseline, DANN, CID-GaitNet.

Ablations:
No causal loss, no adversarial nuisance, no environment grouping.

Expected advantage:
Strong conceptual novelty.

Weakness:
Causal claims must be conservative.

Risk:
High if overclaimed.

Q1 defense:
Frame as causal-inspired invariance, not proof of true causal structure.

Best paper:
Paper 1 or theoretical extension.

A8. Aperture-MoE: Antenna-Aperture Mixture-of-Experts Network

Q1 title:
“Antenna-Aperture-Aware Mixture-of-Experts for Cross-Configuration WiFi CSI Gait Recognition”

Core idea:
Use setup-aware experts:

```
Expert 1: compact regular router
Expert 2: antenna-spread router
Expert 3: shared identity expert
Expert 4: direction/bag robustness expert
```

A gating network selects expert weights from CSI statistics and setup tokens.

Novelty:
Mixture-of-experts is common elsewhere, but router-configuration-aware MoE for CSI gait is likely underexplored.

Input:
CSI tensor with setup label during training. At inference, setup can be known or inferred.

Pipeline:

```
CSI stem
→ shared encoder
→ setup-aware gating
→ expert bank
→ fused embedding
→ identity / domain / condition heads
```

Losses:
ArcFace, load-balancing loss, domain loss, condition loss, consistency loss.

Training:
Pretrain shared expert on regular data; train setup experts on 5 paired subjects; use strong regularization.

Evaluation:
R→M, M→R, mixed-domain ID.

Baselines:
Single encoder, domain-specific BN, DANN, PAIR-BridgeNet.

Ablations:
Number of experts, hard vs soft gating, no load-balancing, no shared expert.

Expected advantage:
Good Paper 2 variant.

Weakness:
500 modified samples may be too few for many experts.

Risk:
Medium-high.

Q1 defense:
Use small expert count and prove parameter efficiency.

Best paper:
Paper 2.

PART 4 — Type B: modified existing architectures with publishable redesigns

These are not “completely new families,” but they can become Q1-level if redesigned around your covariates and hard protocols.

Table 4a — columns: Proposed modified name, Exact modification, Why not trivial, Dataset support, Novelty level

Original architecture	Proposed modified name	Exact modification	Why not trivial	Dataset support	Novelty level
CNN	Subcarrier-Separable CSI-ResCNN	Depthwise separable convolutions over time, subcarrier, antenna-link axes + link attention	CSI axes are treated separately, not as generic image channels	All 25 regular subjects	Medium
CNN-LSTM/GRU	GaitCycle ConvGRU	Conv encoder + GRU with gait-cycle attention and condition-balanced sampling	Uses walking-cycle structure, not generic recurrent pooling	Direction and bag labels	Medium
TCN	Direction-Equivariant TCN	Dilated TCN with latent direction-reversal consistency	Adds explicit L2R/R2L equivariance	Odd/even direction labels	High
ResNet/Inception	Multi-Scale CSI-InceptionNet	Parallel temporal kernels for slow torso vs fast limb motion + subcarrier branch	CSI-specific temporal scale design	Full regular dataset	Medium
Siamese	Covariate-Paired Siamese CSI-Net	Positive pairs are same person across direction/bag; negatives are same condition different person	Pair mining explicitly targets invariance	Full regular dataset	High
Triplet network	Hard-Covariate Triplet GaitNet	Anchor normal-L2R, positive bag-R2L same person, negative same condition different person	Triplets enforce hardest real shifts	Direction + bag labels	High
Vision Transformer	CSI Tubelet ViT	3D tubelets over time × subcarrier × antenna-link, not 2D CSI images	Preserves CSI tensor structure	Amplitude/spectrogram inputs	Medium-high
Time-Series Transformer	PatchTST-CSI	Patch time series per subcarrier/link with shared backbone and cross-channel fusion	Adapts patch time-series idea to CSI axes; PatchTST itself is time-series, not CSI gait	All CSI tensors	Medium-high
Conformer	CSI-Conformer-X	Local convolution for gait micro-dynamics + attention for long-range walking sequence + subcarrier attention	Better inductive bias than vanilla Transformer	25 regular subjects	Q1-potential
Swin Transformer	Shifted CSI-Window Transformer	Shifted windows over time-subcarrier maps with antenna-link tokens	Local-global hierarchy for CSI images	Spectrogram/CSI image	Medium
Graph Neural Network	Antenna-Subcarrier GaitGNN	Graph nodes are subcarrier groups and antenna links; temporal edges added across packets	CSI topology encoded explicitly	Needs stable subcarrier/link metadata	High
Autoencoder	Disentangled CSI Denoising AE	Separate identity/nuisance latent variables; reconstruct CSI while adversarially removing nuisance from identity	Reconstruction + invariance, not simple denoising	All data	Medium-high
Masked autoencoder	Condition-Aware CSI-MAE	Structured masking by packets, subcarriers, antenna links, amplitude/phase; fine-tune with ArcFace	CSI-specific masking strategy	All regular + modified	Q1-potential

Original architecture	Proposed modified name	Exact modification	Why not trivial	Dataset support	Novelty level
Contrastive learning	Multi-Condition SupCon CSI	Positives are same identity across direction/bag/setup; hard negatives share condition	Uses your labels deeply	Full regular + 5 cross-setup	Q1-potential
Domain adaptation	Paired Router DANN++	DANN + paired prototype alignment + setup-specific stem	Uses paired same-subject domains	5 overlap subjects	Q1-potential but data-limited
Multi-task learning	Identity-Nuisance MTLNet	Joint ID, direction, bag, setup heads with gradient reversal from identity branch	MTL plus deconfounding	All labels	High
Lightweight edge model	EdgeCSI-GaitLite	Mobile depthwise CNN + tiny TCN + ArcFace + quantization-aware training	Deployment-aware, not only accuracy	All data	Medium

Table 4b — columns: Difficulty, Expected benefit, Required comparison, Required ablation, Possible contribution statement

Original architecture	Difficulty	Expected benefit	Required comparison	Required ablation	Possible contribution statement
CNN	Easy	Better efficiency and interpretability	Plain CNN, 2D CNN	remove axis separation, remove link attention	“An RF-axis-aware CNN for CSI gait recognition.”
CNN-LSTM/GRU	Medium	Better temporal robustness	CNN-GRU, BiGRU	no gait attention, no balanced sampling	“Condition-aware temporal modeling of CSI gait cycles.”
TCN	Medium	Strong cross-direction performance	TCN, Gate-ID-style augmentation	no equivariance, no cycle loss	“A TCN that models walking direction as a latent transformation.”
ResNet/Inception	Easy-medium	Good baseline-to-contribution bridge	ResNet, InceptionTime	kernel-scale removal	“Multi-scale RF gait feature extraction.”
Siamese	Medium	Strong verification	Siamese, ProtoNet	random pairs vs covariate pairs	“Covariate-aware Siamese learning for CSI biometrics.”
Triplet network	Medium	Better open-set robustness	Triplet CNN, ArcFace	triplet mining variants	“Hard-condition triplet mining for device-free gait biometrics.”
Vision Transformer	Medium	Better than ViT-on-image	ViT, Swin, CNN	patch shape, positional embeddings	“CSI tensor tubelet tokenization for gait recognition.”
Time-Series Transformer	Medium	Efficient long-window modeling	vanilla Transformer, TCN	patch length, channel-independent vs mixed	“Patch-based CSI sequence modeling for gait.”
Conformer	Medium-hard	Strong accuracy/efficiency	Transformer, CNN-GRU, TCN	remove conv module, remove attention	“A Conformer redesigned for CSI gait temporal-local structure.”
Swin Transformer	Hard	Good if CSI image quality high	Swin, ViT, CNN	window size, shift/no shift	“Hierarchical local attention for CSI gait maps.”
Graph Neural Network	Hard	Interpretability and domain robustness	GAT, Transformer	graph topology, random graph	“Graph-structured RF representation learning for gait.”
Autoencoder	Medium	Better small-data representation	AE, denoising AE	no disentanglement, no adversarial	“Nuisance-aware CSI reconstruction for robust biometrics.”
Masked autoencoder	Medium-hard	Strong few-shot performance	MAE, SimCLR, supervised	mask type/ratio	“Masked CSI modeling for few-shot gait recognition.”
Contrastive learning	Medium	Robust embeddings	SimCLR, SupCon, ArcFace	positive-pair policy	“Condition-aware contrastive learning for CSI gait.”

Original architecture	Difficulty	Expected benefit	Required comparison	Required ablation	Possible contribution statement
Domain adaptation	Hard	Better R→M/M→R	DANN, CORAL, MMD	remove pair/prototype losses	“Paired router-configuration domain adaptation.”
Multi-task learning	Medium	Robust features	single-task, MTL no GRL	head/loss removals	“Multi-task deconfounded CSI gait learning.”
Lightweight edge model	Medium	Fast inference	full CNN/Transformer	pruning/quantization	“Robust edge-deployable CSI gait model.”

PART 5 — LLM-inspired architecture analysis

5.1 Is a pure LLM suitable?

No, not as the main classifier. CSI gait recognition is a numerical RF time-series problem. A language model trained on text has no natural prior for subcarriers, phase distortion, antenna links, Fresnel-zone effects, or packet timing. Converting CSI arrays to text and asking an LLM to classify people would look gimmicky.

5.2 Is it valid to tokenize CSI like language?

Yes, but call it tokenized time-series/RF modeling, not an LLM. Tokenization is valid when each token corresponds to a meaningful CSI patch:

```
time patch
subcarrier group
antenna-link group
amplitude/phase modality
Doppler patch
```

This is closer to ViT, PatchTST, BERT-style masked modeling, or MAE than to an LLM. PatchTST and time-series foundation models show that patch/token modeling is serious for numerical sequences, but this does not mean a text LLM is appropriate.

5.3 Can BERT/MAE-style masked CSI modeling be useful?

Yes. This is one of the safest “LLM-inspired” ideas. Masked time-series autoencoders already exist; your novelty would be CSI-specific structured masking and biometric fine-tuning.

5.4 Can GPT-style autoregressive prediction be useful?

Yes, but as self-supervised pretraining, not as a final LLM classifier. Predicting future CSI patches can force the model to learn walking dynamics:

```
given CSI patches 1...t, predict t+1...t+k
```

This may help few-shot and cross-condition recognition.

5.5 Can an LLM be used after CSI model output?

Yes, safely, as a reporting/explanation layer:

```
CSI model outputs:
- predicted identity
- confidence
- direction
- bag condition
- salient subcarriers
- important time regions
- domain-shift score
LLM converts these into a human-readable explanation.
```

But the LLM must not be the scientific core of the paper.

5.6 Serious LLM-inspired architectures

Architecture	Serious idea	Why acceptable
CSI-BERT-MAE	Mask CSI patches and reconstruct amplitude/phase/subcarrier features; fine-tune for ID	Scientifically grounded self-supervised RF representation learning
CSI-GPT-Predictor	Autoregressively predict future CSI patches, then use encoder states for gait ID	Useful for learning walking temporal dynamics
CSI-Tokenizer + Codebook Transformer	Learn a vector-quantized CSI codebook, then model discrete RF tokens	Serious if codebook tokens are RF patches, not text tokens

5.7 Risky LLM ideas to avoid

Risky idea	Why avoid
Convert CSI CSV rows into text and fine-tune an LLM	No meaningful language prior; tiny dataset; reviewer rejection risk
Ask ChatGPT/GPT to classify people from serialized CSI	Not reproducible, not physically grounded
Claim “LLM-based CSI biometric recognition” because you used a Transformer	Misleading; reviewers will see it as trend-chasing

5.8 Safe alternatives

1. Masked CSI Autoencoder instead of “CSI LLM.”
2. PatchTST-style CSI Transformer instead of GPT-style language modeling.
3. Conformer/Transformer encoder with ArcFace instead of text LLM classification.

PART 6 — Best architecture choices for two Q1 papers

Paper 1 recommendation

Strongest architecture

CID-GaitNet + CycleDIR module

Full name:

Condition-Invariant Direction-Equivariant CSI Gait Network

Tentative title

“Condition-Invariant and Direction-Equivariant Representation Learning for WiFi CSI Gait Biometrics”

Central novelty

The model does not merely classify 25 people. It learns:

```
identity representation       → invariant to direction and bag
direction representation      → predictive of L2R/R2L
condition representation      → predictive of normal/bag
direction operator            → maps L2R motion latent to R2L motion latent
```

Why realistic with your dataset

Your 25-person regular-router dataset has enough repeated samples to train this model, and every sample has identity, direction, and condition labels. The bag subset is smaller, but still usable for hard testing and auxiliary training.

Complete model pipeline

```
Raw CSI file
→ amplitude extraction
→ optional phase sanitization
→ train-only normalization
→ windowing
→ CSI tensor X[B,T,S,L,C]
→ CSI stem
→ factorized temporal-subcarrier encoder
→ latent split: z_id, z_dir, z_cond
→ direction-equivariant cycle operator Q
→ ArcFace identity head
→ direction head
→ bag-condition head
→ adversarial nuisance heads on z_id
```

Experiment matrix

Experiment	Train	Test
Closed-set ID	stratified regular samples	held-out regular samples
Cross-direction 1	L2R	R2L
Cross-direction 2	R2L	L2R
Cross-bag	normal	bag
Hard combined	normal L2R	bag R2L
Reverse hard combined	normal R2L	bag L2R
Multi-task	mixed labels	ID + direction + bag
Few-shot	5/10/20 samples/person	remaining
Open-set	20 known persons	5 unknown persons
External stress	regular 5 overlap	modified 5 overlap

Minimum acceptable baselines

1. SVM/RF/XGBoost on statistical/PCA features.
2. 1D CNN.
3. 2D CNN on CSI image/spectrogram.
4. CNN-GRU or CNN-BiGRU.
5. TCN.
6. Vanilla Transformer.
7. Wi-Gait-style multiscale residual Transformer.
8. ArcFace-only CSI encoder.
9. SupCon-only CSI encoder.
10. Gate-ID-inspired direction augmentation baseline.

Ablations

Ablation	Purpose
no direction auxiliary head	prove direction supervision matters
no bag auxiliary head	prove bag supervision matters
no adversarial nuisance removal	prove invariance matters
no orthogonality loss	prove latent separation matters
no CycleDIR module	prove direction equivariance matters

Ablation	Purpose
CE vs ArcFace	prove biometric margin matters
amplitude vs phase vs fused	prove representation choice
random split vs hard split	expose real robustness

Target journals

Best fit:

1. IEEE Internet of Things Journal
2. IEEE Sensors Journal
3. IEEE Transactions on Instrumentation and Measurement
4. Computer Networks
5. Expert Systems with Applications

Paper 2 recommendation

Strongest architecture

PAIR-BridgeNet

Full name:

Paired Antenna-Configuration Invariant Router Bridge Network

Tentative title

“Paired Cross-Configuration Domain Adaptation for WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Setups”

How it uses the 5-person modified-router subset

It uses the 5 overlapping participants as paired domains:

```
regular setup    = source domain
modified setup   = target domain
same subject     = bridge identity
direction/bag    = shared covariates
```

How to frame R→M and M→R

Protocol	Meaning
R→M	Train on regular setup, test on modified setup
M→R	Train on modified setup, test on regular setup
R→M unsupervised DA	labeled regular + unlabeled modified train samples

Protocol	Meaning
R→M few-shot DA	labeled regular + k labeled modified samples/person
M→R few-shot	stress test because modified has fewer subjects

Are 5 participants enough?

For a standalone strong domain-adaptation paper, 5 is weak. For a paired cross-configuration pilot with careful framing, it is usable. Do not claim general antenna-design superiority. Say:

> “We study a paired pilot cross-configuration domain shift between compact and antenna-spread router setups.”

Complete model pipeline

```
Regular CSI branch       Modified CSI branch
↓                         ↓
regular stem             modified stem
↓                         ↓
shared CSI encoder
↓
identity embedding z
↓
prototype memory
↓
domain adversarial head
pair alignment loss
MMD/CORAL loss
ArcFace identity loss
few-shot target adaptation
```

Experiment matrix

Experiment	Source	Target
no adaptation R→M	regular	modified
no adaptation M→R	modified	regular
CORAL R→M	regular	modified
MMD R→M	regular	modified
DANN R→M	regular	modified
PAIR-Bridge R→M	regular + paired bridge	modified
few-shot PAIR	regular + k modified/person	remaining modified
ablation	remove pair/domain/prototype	modified test

Required baselines

CORAL, MMD, DANN, no-adaptation, fine-tuning, domain-specific BN, ArcFace encoder, PAIR-BridgeNet without pair alignment.

Target journals

1. Computer Networks
2. Ad Hoc Networks
3. IEEE Sensors Journal
4. IEEE Internet of Things Journal

PART 7 — Detailed blueprints for top 5 architectures

Blueprint 1 — CID-GaitNet + CycleDIR

Mathematical formulation

Let:

```
X ∈ ℝ^{B×T×S×L×C}
y ∈ {1, …, 25}
d ∈ {0,1}     # direction
c ∈ {0,1}     # condition: normal/bag
```

Encoder:

```
h = E_θ(X)
z_id   = P_id(h)
z_dir  = P_dir(h)
z_cond = P_cond(h)
```

Identity prediction with ArcFace:

```
logit_y = s · cos(θ_y + m)
logit_j = s · cos(θ_j),  j ≠ y
L_arc = CE(logits, y)
```

Supervised contrastive identity loss:

```
L_supcon = ∑_i [ -1/|P(i)| · ∑_{p∈P(i)} log(
    exp(sim(z_i,z_p)/τ) / ∑_{a≠i} exp(sim(z_i,z_a)/τ) ) ]
```

Orthogonality:

```
L_orth = ‖Z_id^T Z_dir‖²_F + ‖Z_id^T Z_cond‖²_F + ‖Z_dir^T Z_cond‖²_F
```

Direction cycle:

```
z_m = P_motion(h)
z_{m,R2L} ≈ Q z_{m,L2R}
Q² ≈ I
L_cycle = ‖z_{m,j} - Q z_{m,i}‖²
L_invol = ‖Q² - I‖²_F
```

Total:

```
L = L_arc
+ λ₁ L_supcon
+ λ₂ CE(H_dir(z_dir), d)
+ λ₃ CE(H_cond(z_cond), c)
+ λ₄ L_adv_dir(z_id)
+ λ₅ L_adv_cond(z_id)
+ λ₆ L_orth
+ λ₇ L_cycle
+ λ₈ L_invol
```

Block diagram explanation

```
CSI tensor
↓
```

```
CSI axis-aware convolutional stem
↓
temporal-subcarrier Transformer/Conformer encoder
↓
shared latent h
├── identity branch z_id → ArcFace + SupCon
├── direction branch z_dir → direction CE
├── condition branch z_cond → bag CE
└── motion branch z_m → CycleDIR direction operator
```

Input tensor shape

```
[B, T, S, L, C]
Example:
B = 32
T = 512 packets/window
S = number of subcarriers
L = antenna links/spatial streams
C = 1 amplitude or 2 amplitude+phase
```

Output tensor shape

```
z_id   = [B, 256]
z_dir  = [B, 64]
z_cond = [B, 64]
identity_logits = [B, 25]
direction_logits = [B, 2]
condition_logits = [B, 2]
```

Layer-by-layer architecture

Block	Layers
Input normalization	train-only z-score
CSI stem	Conv3D/Conv2D separable: (time, subcarrier, link)
Patch embedding	flatten local CSI patches to D=128
Encoder	4–6 Conformer/Transformer blocks
Pooling	attentive statistics pooling
Projection heads	MLPs for identity/direction/condition/motion
Classifiers	ArcFace identity, CE auxiliary heads
Adversarial heads	gradient reversal + direction/bag classifiers
Cycle module	learnable orthogonal-ish matrix Q

Pseudocode

```
x = preprocess_csi(batch)
h_tokens = stem_and_patch_embed(x)
```

```
h = encoder(h_tokens)
h_pool = attentive_pool(h)
z_id = id_projector(h_pool)
z_dir = dir_projector(h_pool)
z_cond = cond_projector(h_pool)
z_motion = motion_projector(h_pool)
id_logits = arcface(z_id, person_id)
dir_logits = dir_head(z_dir)
cond_logits = cond_head(z_cond)
adv_dir_logits = dir_adv_head(grad_reverse(z_id))
adv_cond_logits = cond_adv_head(grad_reverse(z_id))
loss = arcface_loss + supcon_loss + aux_losses + adv_losses + orth_loss + cycle_loss
```

PyTorch-style module structure

```
class CSIAxisStem(nn.Module): ...
class CSIConformerBlock(nn.Module): ...
class AttentiveStatsPool(nn.Module): ...
class ProjectionHead(nn.Module): ...
class ArcFaceHead(nn.Module): ...
class GradientReverseLayer(torch.autograd.Function): ...
class DirectionCycleOperator(nn.Module): ...
class CIDGaitNet(nn.Module): ...
class CIDGaitLoss(nn.Module): ...
```

Training loop overview

```
for epoch:
for batch:
load CSI tensor and labels
forward CIDGaitNet
compute ArcFace/SupCon/aux/adversarial/orth/cycle losses
backward
gradient clipping
optimizer step
validate on grouped split
save checkpoint by hard-split validation metric
```

Hyperparameters

Parameter	Suggested value
embedding dim	128–256
encoder depth	4–6
attention heads	4–8
dropout	0.1–0.3
ArcFace scale	16–32
ArcFace margin	0.2–0.4
SupCon temperature	0.07–0.2

Parameter	Suggested value
batch size	16–64
optimizer	AdamW
learning rate	1e-4 to 3e-4
weight decay	1e-4
epochs	100–200
early stopping	hard validation split

Data augmentation

Use only on training data:

```
packet dropout
subcarrier dropout
mild amplitude scaling
Gaussian jitter
time crop
frequency masking
antenna-link dropout
phase jitter after sanitization
```

Avoid aggressive time reversal unless used only in the direction-equivariance module and validated.

Evaluation metrics

```
accuracy
macro-F1
balanced accuracy
per-person F1
EER
AUROC
TAR@FAR
confusion matrix
mean ± std over seeds
95% CI
Wilcoxon or paired t-test
parameters/FLOPs/latency
```

Expected result table format

Table 13a — columns: Random, Grouped, L2R→R2L, R2L→L2R, Normal→Bag

Model	Random	Grouped	L2R→R2L	R2L→L2R	Normal→Bag

Table 13b — columns: Hard split, Params

Model	Hard split	Params

Ablation table format

Variant	L2R→R2L	Normal→Bag	Hard split	Δ vs full

Reviewer defense points

1. Uses all dataset labels, not just person ID.
2. Hard splits directly test robustness.
3. ArcFace/SupCon gives biometric embedding.
4. Direction is modeled structurally, not only classified.
5. Nuisance information is explicitly separated.

Blueprint 2 — PAIR-BridgeNet

Mathematical formulation

For overlapping subject i:

```
X_i^R = regular setup CSI
X_i^M = modified setup CSI
```

Embeddings:

```
z_i^R = E_shared(E_R(X_i^R))
z_i^M = E_shared(E_M(X_i^M))
```

Pair alignment:

```
L_pair = ‖z_i^R - z_i^M‖²
```

Prototype alignment:

```
p_y^R = mean(z^R | y)
p_y^M = mean(z^M | y)
L_proto = ∑_y ‖p_y^R - p_y^M‖²
```

Domain adversarial loss:

```
L_domain = CE(D(GRL(z)), setup)
```

Total:

```
L = L_arc + λ₁ L_pair + λ₂ L_proto + λ₃ L_domain + λ₄ L_mmd + λ₅ L_supcon
```

Block diagram

```
regular CSI → regular stem  ┐
├→ shared encoder → embedding → ID head
modified CSI → modified stem ┘                  → domain head
→ prototype memory
→ pair alignment
```

Tensor shapes

Input:

```
X_R, X_M ∈ ℝ^{B×T×S×L×C}
```

Output:

```
z ∈ ℝ^{B×256}
id_logits ∈ ℝ^{B×5} for cross-setup subset
domain_logits ∈ ℝ^{B×2}
```

Layer structure

Component	Details
Regular stem	small Conv block
Modified stem	separate Conv block
Shared encoder	TCN/Conformer/Transformer
Prototype memory	class prototypes for 5 overlap subjects
Domain head	gradient reversal classifier
Pair adapter	optional linear residual adapter R→M, M→R

Pseudocode

```
z_r = model.encode_regular(x_regular)
z_m = model.encode_modified(x_modified)
id_logits_r = arcface(z_r, y)
id_logits_m = arcface(z_m, y)
domain_logits = domain_head(grad_reverse(torch.cat([z_r, z_m])))
loss = id_loss + pair_loss(z_r, z_m) + proto_loss + domain_loss + mmd_loss
```

PyTorch-style modules

```
class SetupSpecificStem(nn.Module): ...
class SharedCSIEncoder(nn.Module): ...
class PrototypeMemory(nn.Module): ...
class MMDLoss(nn.Module): ...
class CORALLoss(nn.Module): ...
class PairBridgeNet(nn.Module): ...
class PairBridgeLoss(nn.Module): ...
```

Hyperparameters

Parameter	Value
embedding	128–256
pair loss weight	0.1–1.0
MMD/CORAL weight	0.01–0.2

Parameter	Value
domain loss weight	ramp from 0 to 1
target few-shot k	1, 5, 10, 20
source pretraining	25 regular subjects

Reviewer defense

1. Domain shift is physically meaningful.
2. Same-subject pairing is unique and exploited.
3. Claims are limited to cross-configuration adaptation.
4. Baselines include standard DA methods.

Blueprint 3 — M2CSI-MAE

Mathematical formulation

Masking:

```
M ∈ {0,1}^{T×S×L×C}
X_visible = X ⊙ M
X_masked = X ⊙ (1 - M)
```

Reconstruction:

```
X_hat = D(E(X_visible), M)
L_rec = ‖(X_hat - X) ⊙ (1 - M)‖₁
```

Optional phase derivative:

```
L_phase = ‖Δφ_hat - Δφ‖₁
```

Fine-tuning:

```
L_total = L_arc + λ L_supcon
```

Block diagram

```
CSI tensor
→ structured mask
→ visible patch encoder
→ reconstruction decoder
→ pretraining
then:
encoder
→ ArcFace/SupCon identity head
```

Input/output

```
Input:  [B,T,S,L,C]
Output pretrain: reconstructed [B,T,S,L,C]
Output finetune: identity logits [B,25], embedding [B,256]
```

Layer structure

Block	Details
Patch embedding	time × subcarrier × link patches
Encoder	4–8 Transformer blocks
Decoder	2–4 lightweight Transformer blocks
Reconstruction heads	amplitude, phase, derivative
Fine-tune head	ArcFace

Pseudocode

```
mask = structured_csi_mask(x)
visible = x * mask
latent = encoder(patch_embed(visible, mask))
x_hat = decoder(latent, mask)
pretrain_loss = reconstruction_loss(x_hat, x, mask)
# fine-tune
z = encoder_no_decoder(x)
loss = arcface_loss(z, y) + supcon_loss(z, y)
```

PyTorch modules

```
class StructuredCSIMasker(nn.Module): ...
class CSIPatchEmbed(nn.Module): ...
class MaskedCSIEncoder(nn.Module): ...
class CSIDecoder(nn.Module): ...
class M2CSIMAE(nn.Module): ...
class MAEPretrainLoss(nn.Module): ...
```

Reviewer defense

1. Uses unlabeled CSI efficiently.
2. Masking follows CSI physical axes.
3. Few-shot evaluation proves value.
4. Not presented as an LLM gimmick.

Blueprint 4 — Fresnel-HyperGait

Mathematical formulation

Construct hypergraph:

```
G = (V, E)
v = token(t, s, l)
```

Hyperedges:

```
E_time: same time patch
E_subcarrier: neighboring subcarrier bands
E_link: same antenna link
E_setup: setup-specific spatial grouping
```

Hypergraph attention:

```
h'_v = ∑_{e∋v} α_{v,e} W_e aggregate({h_u | u∈e})
```

Loss:

```
L = L_arc + λ₁ L_supcon + λ₂ L_graph_smooth + λ₃ L_domain
```

Block diagram

```
CSI tensor
→ patch tokens
→ hypergraph construction
→ hypergraph attention
→ temporal pooling
→ identity / condition / setup heads
```

Tensor shapes

```
Input: [B,T,S,L,C]
Tokens: [B,N,D], N = T_p × S_p × L
Output embedding: [B,256]
```

PyTorch modules

```
class CSIHypergraphBuilder(nn.Module): ...
class HypergraphAttentionLayer(nn.Module): ...
class FresnelHyperGaitEncoder(nn.Module): ...
class HypergraphReadout(nn.Module): ...
class FresnelHyperGait(nn.Module): ...
```

Reviewer defense

1. CSI is modeled as RF structure, not image.
2. Antenna-spread configuration is represented explicitly.
3. Hyperedge ablations show which RF relations matter.

Blueprint 5 — ProtoRouter-Gait

Mathematical formulation

Prototype:

```
p_y = mean({z_i | y_i = y})
```

Condition-adapted prototype:

```
p_y^{d,c,s} = A_{d,c,s}(p_y)
```

Query classification:

```
P(y|x) = softmax(-dist(z_x, p_y^{d,c,s}))
```

Open-set score:

```
score(x) = max_y cos(z_x, p_y)
unknown if score < τ
```

Loss:

```
L = L_proto + λ₁ L_triplet + λ₂ L_arc + λ₃ L_open
```

Pipeline

```
support CSI samples
→ encoder
→ identity prototypes
→ direction/bag/setup adapters
→ query classification
→ open-set threshold
```

Tensor shapes

```
Support: [N_way,K,T,S,L,C]
Query: [Q,T,S,L,C]
Embedding: [N_way,K,256], [Q,256]
```

PyTorch modules

```
class EpisodicSampler: ...
class CSIEmbeddingEncoder(nn.Module): ...
class ConditionPrototypeAdapter(nn.Module): ...
class ProtoRouterClassifier(nn.Module): ...
class OpenSetThresholdCalibrator: ...
```

Reviewer defense

1. Realistic biometric enrollment.
2. Directly evaluates 5/10/20-shot recognition.
3. Better than closed-set-only classification.

PART 8 — Novelty and publishability scoring

Scores are reviewer-style estimates, not guarantees.

Table 18a — columns: Architecture, Type, Novelty, Depth, Dataset fit

Rank	Architecture	Type	Novelty	Depth	Dataset fit
1	CID-GaitNet + CycleDIR	New	9	9	10
2	PAIR-BridgeNet	New	9	9	7
3	M2CSI-MAE	New/modified SSL	8	9	9
4	ProtoRouter-Gait	New	8	8	9
5	Fresnel-HyperGait	New	9	9	7
6	CSI-Conformer-X	Modified	7	8	9
7	Hard-Covariate Triplet GaitNet	Modified	7	8	10
8	Multi-Condition SupCon CSI	Modified	7	8	10
9	CSI-CausalGait	New	9	9	8
10	Aperture-MoE	New	8	8	6
11	PatchTST-CSI	Modified	6	7	8
12	Antenna-Subcarrier GaitGNN	Modified	8	8	7
13	Condition-Aware CSI-MAE	Modified	7	8	9
14	Direction-Equivariant TCN	Modified	7	7	10
15	Covariate-Paired Siamese CSI-Net	Modified	6	7	9
16	CSI Tubelet ViT	Modified	6	7	8
17	Subcarrier-Separable CSI-ResCNN	Modified	5	6	9
18	GaitCycle ConvGRU	Modified	5	6	9
19	Multi-Scale CSI-InceptionNet	Modified	5	6	8
20	Shifted CSI-Window Transformer	Modified	6	7	7
21	EdgeCSI-GaitLite	Modified	5	6	8
22	CSI-GPT-Predictor	LLM-inspired	6	7	7
23	CSI Codebook Transformer	LLM-inspired	7	8	7
24	Pure LLM CSI classifier	LLM	2	2	2

Table 18b — columns: Feasibility, Risk, Q1 publishability, Verdict

Rank	Feasibility	Risk	Q1 publishability	Verdict
1	8	4	9	Implement first
2	7	7	8	Best Paper 2, data-limited
3	8	5	8.5	Strong if few-shot gains appear

Rank	Feasibility	Risk	Q1 publishability	Verdict
4	8	5	8	Strong biometric framing
5	6	8	7.5	Novel but harder
6	8	4	7.5	Strong backbone
7	8	4	7.5	Good open-set model
8	9	3	7.5	Excellent practical choice
9	6	8	7	Strong but risky
10	7	7	7	Save for Paper 2
11	8	4	6.5	Good baseline-plus
12	6	7	6.5	Needs metadata
13	8	5	7.5	Close to M2CSI-MAE
14	8	4	7	Good ablation/backbone
15	8	4	6.5	Good verification baseline
16	7	5	6	Needs more than tokenization
17	9	3	6	Strong baseline, not main novelty
18	8	3	5.5	Baseline-level
19	9	3	5.5	Good baseline
20	6	6	5.5	Risk of looking like Swin copy
21	9	3	5.5	Good secondary contribution
22	6	7	5.5	Use only as SSL auxiliary
23	5	8	5.5	Interesting but risky
24	4	10	1	Avoid

Too weak

1. Plain CNN-LSTM.
2. Plain Transformer.
3. Plain ViT on spectrogram.
4. Simple attention CNN.
5. Softmax-only classifier.

Too risky

1. Pure LLM classifier.
2. Large MoE with only 500 modified samples.
3. Strong causal claims without careful protocol.

4. Diffusion/GAN counterfactual generation as the main novelty.

Publishable but need more data

1. Aperture-MoE.
2. Fresnel-HyperGait.
3. PAIR-BridgeNet as a standalone antenna-configuration paper.
4. Cross-setup open-set biometric authentication.

Realistic with current data

1. CID-GaitNet + CycleDIR.
2. M2CSI-MAE.
3. Multi-Condition SupCon CSI.
4. ProtoRouter-Gait.
5. Hard-Covariate Triplet GaitNet.
6. CSI-Conformer-X.

PART 9 — Mandatory experimental protocol for selected best architecture

Selected model:

CID-GaitNet + CycleDIR

9.1 Subject identification

```
Task: 25-class closed-set person ID
Data: regular-router setup
Split: grouped sample split, not only random split
Metrics: accuracy, macro-F1, balanced accuracy, per-person F1
```

9.2 Direction robustness

Protocol	Train	Test
D1	L2R samples	R2L samples
D2	R2L samples	L2R samples

Report performance drop relative to mixed-direction training.

9.3 Bag robustness

Protocol	Train	Test
B1	samples 1–80 normal	samples 81–100 bag
B2	normal L2R	bag R2L
B3	normal R2L	bag L2R

9.4 Multi-condition recognition

Multi-task outputs:

```
person_id: 25 classes
direction: 2 classes
condition: 2 classes
```

Report:

```
ID accuracy
direction accuracy
bag accuracy
macro-F1 for all tasks
```

9.5 Cross-setup transfer

Use the 5 overlapping subjects:

Protocol	Train	Test
R→M	regular 5 subjects	modified 5 subjects
M→R	modified 5 subjects	regular 5 subjects
R→M few-shot	regular + k modified/person	remaining modified
M→R few-shot	modified + k regular/person	remaining regular

9.6 Few-shot recognition

For 25 regular subjects:

```
5-shot/person
10-shot/person
20-shot/person
```

Use remaining samples as queries. Report mean ± std over 20 episodic runs.

9.7 Open-set recognition

Example:

```
Known identities: 20 participants
Unknown identities: 5 participants
Enrollment: 5/10/20 samples per known person
Query: remaining known + all unknown
Metrics: EER, AUROC, TAR@FAR=1%, TAR@FAR=5%
```

9.8 Self-supervised pretraining

Optional but valuable:

```
Pretrain M2CSI-MAE on all regular + modified CSI without labels
Fine-tune CID-GaitNet on labeled regular data
Compare with training from scratch
```

9.9 Statistical testing

For every major result:

```
5 or 10 random seeds
mean ± standard deviation
95% confidence interval
paired t-test or Wilcoxon signed-rank test
McNemar test for paired classifier comparison
effect size
```

9.10 Complexity

Report:

```
parameter count
```

```
FLOPs
inference latency per sample
GPU/CPU memory
window length sensitivity
subcarrier reduction sensitivity
```

PART 10 — Final recommendations

10.1 Top 10 architecture ideas

1. CID-GaitNet + CycleDIR
2. PAIR-BridgeNet
3. M2CSI-MAE
4. ProtoRouter-Gait
5. Fresnel-HyperGait
6. CSI-Conformer-X
7. Hard-Covariate Triplet GaitNet
8. Multi-Condition SupCon CSI
9. CSI-CausalGait
10. Aperture-MoE

10.2 Best 3 completely new architectures

1. CID-GaitNet + CycleDIR — best for Paper 1.
2. PAIR-BridgeNet — best for Paper 2.
3. Fresnel-HyperGait — highest novelty, but harder and more metadata-dependent.

10.3 Best 3 modified architectures

1. CSI-Conformer-X
2. Condition-Aware CSI-MAE
3. Hard-Covariate Triplet GaitNet

10.4 Best 2 LLM-inspired but scientifically safe architectures

1. M2CSI-MAE / CSI-BERT-MAE
2. CSI-GPT-Predictor used only for self-supervised future-patch prediction

Do not call these “LLM models” in the title. Call them masked CSI modeling, autoregressive CSI pretraining, or self-supervised RF sequence modeling.

10.5 Final Paper 1 architecture

CID-GaitNet + CycleDIR

Paper title:

“Condition-Invariant and Direction-Equivariant Representation Learning for WiFi CSI Gait Biometrics”

10.6 Final Paper 2 architecture

PAIR-BridgeNet

Paper title:

“Paired Cross-Configuration Domain Adaptation for WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Setups”

10.7 Complete implementation order

1. Build metadata table and leakage-safe split files.
2. Implement amplitude-only preprocessing.
3. Implement classical ML baselines.
4. Implement CNN, CNN-GRU, TCN, vanilla Transformer.
5. Implement ArcFace encoder.
6. Implement SupCon identity encoder.
7. Implement CID-GaitNet without CycleDIR.
8. Add adversarial direction/bag removal.
9. Add CycleDIR operator.
10. Add phase branch only after amplitude pipeline is stable.
11. Implement few-shot/open-set evaluation.
12. Implement PAIR-BridgeNet for regular↔modified transfer.
13. Add M2CSI-MAE pretraining if time allows.

10.8 Exact first experiment to run

Run this first:

```
Amplitude-only 25-person regular-router closed-set ID
Models:
1. SVM on statistical/PCA features
2. 1D CNN
3. TCN
4. ArcFace CSI encoder
Splits:
1. grouped sample split
2. L2R→R2L
3. normal→bag
```

This tells you whether the dataset has strong identity signal under hard conditions before building complex models.

10.9 Exact baselines you must implement

Minimum baseline list:

```
SVM
```

```
Random Forest or XGBoost
1D CNN
2D CNN on CSI image/spectrogram
CNN-GRU or CNN-BiGRU
TCN
vanilla Transformer
CSI-Conformer baseline
ArcFace encoder
SupCon encoder
DANN/CORAL/MMD for cross-setup
ProtoNet/Siamese/Triplet for few-shot/open-set
```

10.10 Exact ablations you must include

For CID-GaitNet:

```
Full model
minus direction head
minus bag head
minus adversarial nuisance heads
minus orthogonality loss
minus supervised contrastive loss
minus ArcFace, CE only
minus CycleDIR
amplitude only
phase only
amplitude + phase
no augmentation
no hard positive mining
```

For PAIR-BridgeNet:

```
Full model
minus pair alignment
minus domain adversarial loss
minus MMD/CORAL
minus prototype memory
minus setup-specific stems
regular-only pretraining vs no pretraining
few-shot k = 1,5,10,20
```

Complete PyTorch implementation plan for the best architecture

Best architecture:

CID-GaitNet + CycleDIR

A. Project structure

```
csi_gait/
configs/
cid_gaitnet.yaml
baselines.yaml
```

```
pair_bridge.yaml
data/
raw/
processed/
splits/
metadata_master.csv
csi/
readers/
picoscenes_reader.py
preprocessing/
amplitude.py
phase_sanitization.py
filters.py
windowing.py
normalization.py
datasets/
csi_dataset.py
episodic_dataset.py
split_protocols.py
models/
stems.py
encoders.py
pooling.py
heads.py
losses.py
cid_gaitnet.py
pair_bridge_net.py
mae.py
train/
train_cid.py
train_baselines.py
train_mae.py
train_pair_bridge.py
eval/
evaluate_closed_set.py
evaluate_cross_direction.py
evaluate_cross_bag.py
evaluate_fewshot.py
evaluate_openset.py
evaluate_cross_setup.py
logs/
tensorboard/
wandb/
csv/
results/
tables/
figures/
checkpoints/
```

B. Dataset output

Each __getitem__ should return:

```
{
"x": Tensor[T, S, L, C],
"person_id": LongTensor[],
"direction": LongTensor[],   # 0=L2R, 1=R2L
"condition": LongTensor[],   # 0=normal, 1=bag
"setup": LongTensor[],       # 0=regular, 1=modified
"sample_id": int,
"participant_id": str
}
```

Batch shape:

```
x: [B, T, S, L, C]
person_id: [B]
direction: [B]
condition: [B]
setup: [B]
```

C. Main modules

```
class CSIAxisStem(nn.Module):
"""
Input:  [B, T, S, L, C]
Output: [B, N, D]
"""
```

Suggested operations:

```
permute to [B, C, T, S, L]
Conv3D(C → 32, kernel=(7,3,1))
BatchNorm / GroupNorm
GELU
Conv3D depthwise over time/subcarrier
1×1×1 projection to D
patch flatten to tokens
class CSIConformerEncoder(nn.Module):
"""
Input:  [B, N, D]
Output: [B, N, D]
"""
```

Each block:

```
LayerNorm
Multi-head self-attention
residual
LayerNorm
depthwise temporal convolution module
residual
Feed-forward network
residual
class AttentiveStatsPool(nn.Module):
"""
Input:  [B, N, D]
Output: [B, 2D]  # weighted mean + weighted std
"""
```

```
class ProjectionHead(nn.Module):
"""
MLP: input_dim → hidden_dim → output_dim
"""
class ArcFaceHead(nn.Module):
"""
Input: embedding [B, E], labels [B]
Output: logits [B, num_classes]
"""
class GradientReverseLayer(Function):
"""
Used for adversarial direction/condition classifiers.
"""
class DirectionCycleOperator(nn.Module):
"""
Learn Q ∈ ℝ^{E_m×E_m}
Enforces Q² ≈ I.
"""
class CIDGaitNet(nn.Module):
def __init__(
self,
num_identities=25,
input_modalities=1,
embed_dim=128,
id_dim=256,
nuisance_dim=64,
motion_dim=128,
depth=4,
heads=4,
):
...
```

Forward output:

```
{
"z_id": z_id,                        # [B,256]
"z_dir": z_dir,                      # [B,64]
"z_cond": z_cond,                    # [B,64]
"z_motion": z_motion,                # [B,128]
"id_logits": id_logits,              # [B,25]
"dir_logits": dir_logits,            # [B,2]
"cond_logits": cond_logits,          # [B,2]
"adv_dir_logits": adv_dir_logits,    # [B,2]
"adv_cond_logits": adv_cond_logits,  # [B,2]
}
```

D. Loss functions

```
class CIDGaitLoss(nn.Module):
def forward(outputs, labels, pair_indices=None):
L_id = arcface_ce(outputs["id_logits"], person_id)
L_supcon = supervised_contrastive(outputs["z_id"], person_id)
L_dir = ce(outputs["dir_logits"], direction)
L_cond = ce(outputs["cond_logits"], condition)
```

```
L_adv_dir = ce(outputs["adv_dir_logits"], direction)
L_adv_cond = ce(outputs["adv_cond_logits"], condition)
L_orth = orthogonality(outputs["z_id"], outputs["z_dir"], outputs["z_cond"])
L_cycle = direction_cycle_loss(outputs["z_motion"], direction, person_id, condition)
L_invol = ||Q @ Q - I||_F^2
return weighted_sum(...)
```

Recommended weights:

```
loss:
arcface: 1.0
supcon: 0.2
direction_ce: 0.2
condition_ce: 0.2
adversarial_direction: 0.1
adversarial_condition: 0.1
orthogonality: 0.05
cycle: 0.1
involution: 0.01
```

E. Training schedule

```
optimizer: AdamW
lr: 0.0003
weight_decay: 0.0001
batch_size: 32
epochs: 150
warmup_epochs: 10
scheduler: cosine
gradient_clip: 1.0
dropout: 0.2
early_stopping_metric: hard_split_macro_f1
seeds: [1, 2, 3, 4, 5]
```

Training stages:

```
Stage 1: train ArcFace encoder only
Stage 2: add direction and bag auxiliary heads
Stage 3: add adversarial nuisance heads
Stage 4: add CycleDIR loss
Stage 5: optional amplitude+phase fusion
```

F. Experiment logging

Log every run with:

```
run_id
git_commit
config_hash
seed
split_name
model_name
input_type
window_length
normalization_method
```

```
accuracy
macro_f1
balanced_accuracy
eer
auroc
params
flops
latency_ms
best_epoch
confusion_matrix_path
embedding_umap_path
checkpoint_path
```

Use:

```
TensorBoard or Weights & Biases
CSV summary for paper tables
JSON config for reproducibility
```

G. First implementation milestone

The first working milestone should be:

```
CID-GaitNet without CycleDIR
amplitude-only
25-person regular setup
grouped split + L2R→R2L + normal→bag
```

Do not add phase, MAE, domain adaptation, or modified-router experiments until this milestone is stable.

Final Q1 publication strategy

1. **Paper 1 first:** CID-GaitNet + CycleDIR on the 25-person regular-router dataset.
2. **Main claim:** robust CSI gait biometrics under direction and carrying-condition shifts.
3. Do not rely on random accuracy. Make cross-direction and normal→bag the main story.
4. **Paper 2 second:** PAIR-BridgeNet for regular↔modified router adaptation.
5. **Frame modified setup carefully:** paired pilot cross-configuration domain shift, not universal antenna-array superiority.
6. Collect more modified-router subjects if possible. Increasing from 5 to 15–25 would greatly strengthen Paper 2.
7. Release split files and metadata. This directly addresses leakage and reproducibility criticism.
8. Use LLM-inspired methods only as masked/autoregressive CSI pretraining. Do not market the paper as LLM-based CSI recognition.
9. **Your strongest publishable story:** not “high CSI classification accuracy,” but architecture-level robust representation learning that explains and reduces failures caused by direction, bag carrying, and router-configuration shift.